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Synergistic effects of rice husk ash and extracted microsilica on the performance of high-strength concrete

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The construction industry's urgent need for sustainable alternatives to ordinary Portland cement (OPC) has spurred significant interest in supplementary cementitious materials (SCMs). Therefore, this study investigates the synergistic effects of rice husk ash (RHA) and extracted microsilica (EMS) as partial replacements for cement in high-strength concrete (HSC). RHA was produced by controlled combustion of rice husk (RH) at 600–700 °C, while EMS was obtained through an optimized extraction process. Along with control mix (CM), three ternary mixes incorporating 15–35% RHA and 5–15% EMS were investigated to evaluate their effects on the mechanical and durability properties, as well as microstructural characteristics. Experimental results demonstrated that samples containing 20% (5% EMS + 15% RHA) and 35% (10% EMS + 25% RHA) as cement replacement exhibited enhanced performance, increasing the 28-day compressive strength by 11.56% and 5.98%, respectively, as compared to conventional concrete. Durability assessments of the optimal mixes revealed a significant reduction in water absorption (WA) and permeability. Physiochemical characterization confirmed the increased formation of dense calcium silicate hydrate (CSH) gels and a reduction in portlandite content, indicating high pozzolanic reactivity. However, the 50% replacement mix (15% EMS + 35% RHA) showed inferior performance due to the particle agglomeration and higher porosity. This study shows that the strategically formulated combinations of EMS and RHA blends (up to 35%) can produce sustainable HSC with improved mechanical properties and durability while mitigating environmental impacts. These findings provide valuable insights for advancing eco-friendly construction materials, supporting global sustainability initiatives in infrastructure development.

Keywords Rice husk ash, Extracted microsilica, Mechanical strength, Durability, Water absorption

Concrete, a composite blend of synthetic and natural constituents mixed with water, is considered the most substantial, versatile, durable, and widely utilized construction material worldwide. As a fundamental construction material, concrete is vital to the growth of a nation's infrastructure and construction industry, contributing to its economic growth and social advancement. Due to rapid expansion in infrastructure development, concrete use continues to grow¹. The global production of concrete is approximately 4.4 billion tons annually, and this number will increase to 5.5 billion tons by 2050, driven by the rapid urbanization in developing countries². To achieve sustainability and effectively reduce carbon emissions, the cement and concrete manufacturing industries must improve process efficiency by incorporating waste materials into OPC blends as partial or full replacements³. According to the report of the International Energy Agency, the carbon dioxide (CO₂) emissions intensity from the cement industry increased by almost 10% since 2015, highlighting the urgent need to use alternative raw materials in cement production⁴. Cement production has historically left a significant carbon footprint, prompting researchers to explore industrial byproducts as SCMs. In recent years, RHA has been extensively examined for its potential use as an SCM⁵. The increasing demand for cement has also led to a greater reliance on raw materials, particularly limestone, which is essential for cement production⁶. In the cement manufacturing process, the calcination reaction produces calcium oxide, releasing CO₂ as a byproduct, while an additional 50%

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of emissions are linked to energy consumption during production⁷. The cement industry produces nearly 5% to 7% of the world's CO₂ emissions, with CO₂ accounting for 65% of total greenhouse gas emissions worldwide⁸. To mitigate the carbon footprint of cement manufacturing, researchers are investigating the potential of industrial byproducts, such as SCMs⁹. Environmental issues associated with the construction industry can be addressed through sustainable approaches, such as substituting cement with highly pozzolanic SCMs to produce durable, high-performance, and sustainable concrete¹⁰.

In recent decades, pozzolanic materials obtained from agricultural and industrial wastes have been widely utilized as SCMs for partial replacement of cement. The inclusion of pozzolanic materials such as ground granulated blast furnace slag (GGBS), fly ash (FA), RHA, and silica fume (SF) refines the concrete microstructure, leading to superior mechanical performance and enhanced durability¹¹. The integration of agricultural residues such as RHA, corncob ash, and sugarcane bagasse ash, along with industrial by-products like FA, GGBS, and metakaolin (MK), has shown favorable outcomes in producing sustainable, high-performance concrete, enhancing strength while improving cost, energy, and environmental efficiency¹². Moreover, using such materials presents sustainable solutions for managing industrial and agricultural waste, particularly in developing countries, where disposing of agro-waste remains a critical challenge¹³. Furthermore, the construction industry has increasingly explored agricultural waste materials, including RHA and bagasse ash, as sustainable and high-performing partial substitutes for cement in concrete production¹⁴. RH is a major agrarian byproduct, with a global annual production of approximately 150 million tons¹⁵. Despite this, only 17% of RHA is utilized for fertilizer, fuel, or electricity generation, while the remaining 83% is discarded or openly burned, causing water and soil pollution¹⁶. Rice, a major Kharif crop in Pakistan, plays a vital role as both a staple food and a key export commodity, contributing 3.1% value added in agriculture and 0.6% to the national GDP¹⁷. According to the literature, 20% of rice paddy is husk, with Pakistan generating approximately 1.5 million tons of RH annually. In Khyber Pakhtunkhwa province, the estimated annual rice production is approximately 147,000 tons, resulting in 30,000 tons of RH¹⁸.

RH is an agricultural waste material with low nutrient content, undergoes significant transformation when burned under controlled conditions, and produces amorphous silica with large surface area. The utilization of RHA as SCM has gained considerable attention in the building construction industry, particularly in rice-producing countries such as Pakistan⁵. RHA, primarily composed of non-crystalline (amorphous) SiO₂, exhibits high pozzolanic activity and enhances the mechanical strength and durability performance of concrete when utilized as a substitute for cement¹⁹. Due to its economic and environmental benefits, RHA is increasingly being explored as an alternative to conventional cementitious materials^{20,21}. Studies show that the combustion of RH at controlled temperatures generates RHA with a high silica content, typically varying between 90% and 95% depending upon the burning conditions. The pozzolanic reactivity of RHA is significantly influenced by factors including incineration temperature, duration, and oxygen availability²². The presence of unburnt carbon in RHA has been observed to impair its pozzolanic reactivity significantly, underscoring the necessity of implementing controlled incineration processes to optimize its performance as an SCM²³. RHA contains amorphous silica, and minimal unburnt carbon enhances concrete properties, such as mechanical strength, reduced porosity, and improved durability²⁴. High-quality pozzolanic materials are produced by combusting RH under controlled conditions, improving the properties of mortar and concrete²⁵. Previous studies indicate that burning RHA at temperatures below 700 °C produces a high percentage of amorphous silica, with optimal pozzolanic reactivity achieved between 500 and 700 °C¹⁵. Research findings suggest that burning RH at 700 °C yields the highest amorphous silica content, accompanied by high pozzolanic reactivity and improved surface area²⁰. A ferrocement furnace with a capacity of 60 kg was utilized to obtain high-quality RHA. Initially, a controlled fire source was maintained for approximately 10 min; after that, the husks gradually combusted for over a day, maintaining a temperature range of 600 to 700 °C. The ash was allowed to cool before being carefully collected, resulting in RHA with 87% amorphous silica²⁶. Several studies have used ferrocement furnaces to obtain high-quality RH ashes with varying unburnt carbon content and a high silica content, ranging between 85% and 97%²⁷.

Concrete production has improved by incorporating RHA as a replacement for SF, reducing environmental pollution and lowering costs. Previous study demonstrated that RHA improves concrete's mechanical and microstructural properties while also providing excellent durability, high frost resistance, and chemical resistance²⁸. The incorporation of RHA improved compressive strength due to its finer particle size and higher silicon-to-aluminum (Si/Al) ratio compared to GGBS²⁹. Due to its pozzolanic nature, RHA has been blended into cement at levels of up to 30% to investigate its impact on consistency, setting time, and permeability, yielding improved results³⁰. Concrete containing 15% or more of RHA significantly enhance compressive strength, demonstrating that reactive silica content is more critical than FA for developing high-performance geopolymers³¹. Earlier studies confirm that amorphous silica refines the interfacial transition zone, densifying the concrete microstructure. RHA incorporation improves hardened properties, with research determining optimal replacement levels (typically 10–20%), curing effects, and strength-resistivity relationships³². Premkumar et al.³³ investigated the combined effect of RHA and SF as partial cement replacements enhances the mechanical and durability properties of concrete, reducing cement consumption, and promoting sustainable construction practices. Finely ground RHA has demonstrated pozzolanic reactivity and durability performance comparable to silica fumes, despite its lower silica content and larger particle size, due to its high specific surface area and porous structure³⁴. RHA demonstrated durability and strength enhancements comparable to SF, with microstructural densification contributing to improved performance, highlighting its cost-effective potential for sustainable concrete in rice-producing regions³⁵. Before advancements in concrete technology, RHA was frequently discarded in landfills and lakes, contaminating the environment³⁶. The advancements in green materials enhanced the utilization of RHA in mortar and concrete, enabling researchers to replace cement with RHA without compromising the mechanical properties³⁷. In the ferrosilicon industry, SF is generally obtained as a by-product in electric arc furnaces at a relatively high temperature of approximately 2000 °C³⁸. Considering

cement and other related industries, silica extraction from various inorganic sources, including silica sand, diorite sand, and waste-colored glasses, has received considerable attention. Based on literature, RHA calcination (600–700 °C) likely has a lower carbon footprint than SF production (2000 °C) due to biomass carbon neutrality and avoided industrial processing³⁹, though chemical treatment steps require optimization. A substitution of RHA resulted in a reduction in the carbon footprint (global warming potential) compared to the use of silica fume, making it eco-friendly⁴⁰. RHA was sourced locally from Charsadda, Khyber Pakhtunkhwa, and was combusted and ground to ash, achieving 100% passing at 75 µm particle size⁵. The emission factors are 0.1032 kg CO₂-eq/kg for combustion⁴¹ and 0.011 kg CO₂-eq/kg for grinding⁴². In contrast, the production of silica fumes is associated with a higher CO₂ emissions footprint compared to that of RHA, primarily due to the elevated temperatures required during its manufacturing process⁴³. In contrast, silica extraction from various inorganic sources often requires complex processing techniques, including crushing, grinding, and high-energy milling, followed by impurity purification. In some cases, high-temperature sintering is also needed⁴⁴. In contrast, silica extracted from well-known biological sources such as sugarcane bagasse ash and RHA are of superior quality and energy efficient. The amorphous and chemically reactive silica extracted from RHA exhibits high purity levels, significant chemical reactivity, and homogeneity^{45,46}. Researchers have evaluated the sustainability of several RHA applications, including microsilica (MS) extracted from RHA⁴⁷. A promising alternative to SF could be the EMS, which offers both economic and environmental advantages⁴⁸.

Despite the high purity, high surface area, pozzolanic reactivity, and homogeneity of extracted silica from RHA, studies exploring its potential applications in cement, mortar, and concrete remain limited. Considering these properties, further exploration is needed to evaluate their full potential in cement-based materials. The partial cement replacement with EMS at low dosages (5% and 15%) significantly improved the mechanical and microstructural performance of mortar while also mitigating alkali-silica expansion⁴⁸. Cavalcante⁴⁹ reported that 16% silica extracted from RHA enhanced cement paste strength at all ages, surpassing 40% MK blends by 28 days. The improvement was attributed to the silica's high reactivity and ultra-fine particles, which densified the microstructure and accelerated hydration. Microstructural analysis confirmed its effectiveness in refining pores and boosting mechanical performance. According to the modified Chapelle test, silica gel extracted from RHA contributed to the consumption of calcium hydroxide (CH), further confirming its amorphous nature. A 5% substitution of cement with extracted silica gel resulted in significantly enhanced compressive strength compared to control cement for the same curing age⁴⁶. To optimize the performance and address compatibility issues of RHA, incorporating additional SCMs with higher pozzolanic activity and high surface area, such as MS, is recommended⁵⁰.

The rapid advancements in infrastructure development have increased the demand for high-resilience and HSC, focusing on a long-term goal of creating sustainable and high-quality SCMs. Therefore, this study aims to explore the combined use of RHA and EMS as high-volume (20–50%) cement replacements for developing sustainable concrete. The specific goals were to assess the mechanical performance and durability of ternary concrete mixes and to investigate their microstructural and pore structure modifications using advanced characterization techniques, highlighting the potential of RHA and EMS as viable alternatives to conventional silica-based materials in HSC.

Research significance

In response to the growing need for low-carbon construction materials, this study explores the innovative application of regionally sourced RHA and EMS as partial replacements for cement to produce more sustainable concrete. While prior research has often explored the individual effects of RHA or MS on concrete, less attention has been paid to their synergistic application, especially when using locally sourced materials and extracted silica from agricultural waste. The current study addresses this gap by evaluating the mechanical performance, durability, and microstructural characteristics of concrete incorporating varying high dosages of RHA (15%, 25%, and 35%) and EMS (5%, 10%, and 15%). The mechanical strength, durability, and microstructural behavior were examined at multiple curing ages (14, 28, and 56 days). X-ray Diffraction (XRD), X-ray fluorescence (XRF), Scanning Electron Microscopy with Energy Dispersive Spectroscopy (SEM-EDS), Fourier Transform Infrared Spectroscopy (FTIR), and nitrogen (N₂) adsorption analysis were used to understand hydration processes, pore structure, and silica reactivity. This research holds potential to influence future studies by establishing a foundation for optimized ternary binder systems using agricultural waste. Furthermore, this study promotes sustainable construction practices, supports policy efforts to reduce cement-related carbon emissions, and offers a viable, cost-effective alternative to commercial SF, especially relevant for regions with abundant biomass resources. Ultimately, the research aims to promote the industrial use of regionally available RHA as a substitute for commercial pozzolans in the production of cost-effective and high-performance sustainable concrete.

Experimental investigation

Materials

Cement and aggregates

Ordinary Portland Cement type-1 from Bestway Cement, Pakistan, was utilized as per ASTM C-150⁵¹. Locally sourced fine and coarse aggregates were utilized as filler materials in concrete mixtures. The gradation curves following ASTM C136⁵² are displayed in Fig. 1.

The coarse aggregate, comprised of 62% (20 mm nominal size) and 38% (10 mm nominal size), was selected to achieve optimal particle packing density and workability, as determined through preliminary mix trials and following established mix design guidelines⁵³. The ratio ensures structural stability (provided by larger-size aggregates) while improving compaction (facilitated by smaller-size aggregates), thereby improving both mechanical performance and practical application. As per ASTM C128⁵⁴, the fine aggregates had a specific

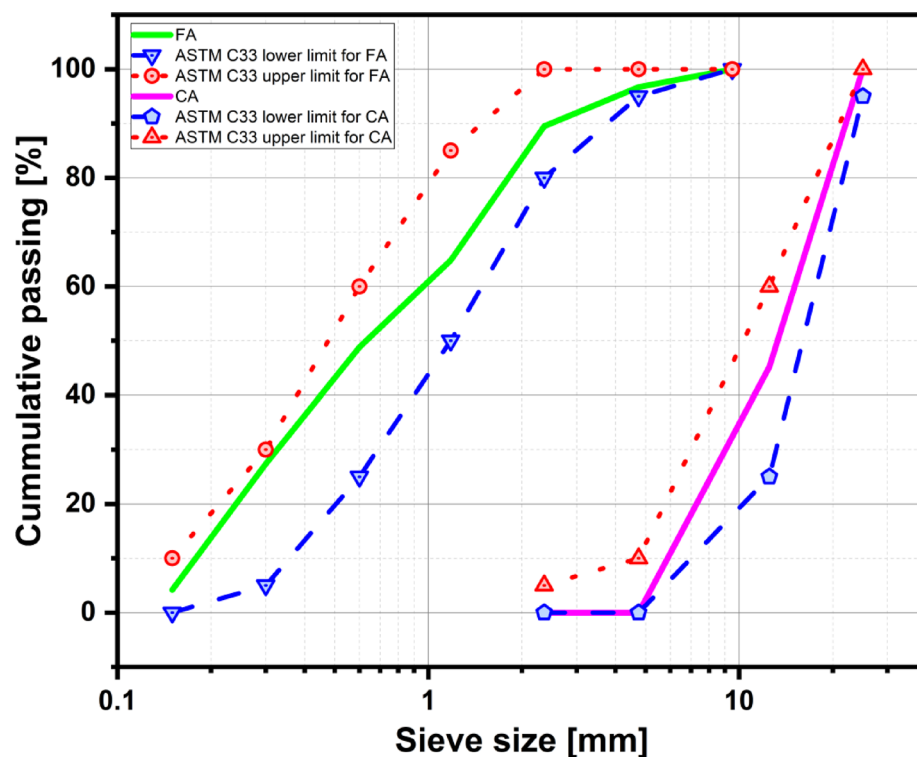


Fig. 1. Gradation curves for aggregates.

gravity, water absorption and fineness modulus of 2.57, 1.78%, and 2.61 respectively. In compliance with ASTM C127⁵⁵, the coarse aggregate exhibited a specific gravity and water absorption of 2.63 and 0.79%, respectively.

Rice husk Ash

Collection of rice husk RH, a byproduct of rice milling, is generated in significant quantities globally as rice serves as a staple food for over half of the global population, making it the second most widely produced cereal crop globally⁵⁶. In 2023, the world's milled rice production was approximately 513 million metric tons, equivalent to approximately 102 million metric tons of RH⁵⁷. China produced 144.62 million rice in 2023, with the highest production globally, as shown in Fig. 2⁵⁸. In Pakistan, rice is cultivated on 2.883 million hectares, and the annual production of rice is 7.32 million tons, as mentioned in Table 1⁵⁹. The chemical composition of RHA varies based on its source, organic composition, and incineration method, influenced by regional meteorological and geographical conditions. Researchers observed significant differences in RHA composition across regions. RH samples were obtained from Charsadda, Khyber Pakhtunkhwa, due to their favorable climate and geography, which contributed to a higher silica content than other regions of the province.

Soil properties Soil plays a pivotal role in plant and crop growth and productivity. Soil is influential in producing healthy plants and proper crop growth. Several significant elements and their oxides are found in soil, such as silicon (Si), calcium (Ca), magnesium (Mg), potassium (K), manganese (Mn), phosphorus (P), sodium (Na), aluminum (Al), and iron (Fe) separated from parent source generally covered by igneous or sedimentary rocks which greatly influenced the production of crops. In addition, silica (SiO_2) is identified as a critical part of the soil profile and a fundamental element of plant cell walls, which substantially influences the physical and chemical characteristics of soil and crops. Moreover, Alumina contributes to physicochemical soil properties, and iron oxide contributes to soil fertility; additionally, other oxides, CaO , Na_2O , MgO , and K_2O , are also present in trace amounts, affecting soil properties⁵. As per the literature, RH samples were collected from Charsadda, Khyber Pakhtunkhwa, based on soil properties having the maximum amount of Silica as a major oxide. The properties and major oxide composition for the selected regions are illustrated in Table 2.

Production of RHA In the current study, the RH was burned at a controlled temperature of 600–700 °C using a ferroceement drum for 24 h to produce high silica RHA. Subsequently, the burning of RH produced approximately 22% of the ash. After the incineration, the ash gradually turned cool and was left in the furnace before being ground in a rotary mill at 15 rpm for 12 h to obtain finer material. After grinding, the sample was packed in polythene bags for further investigation. The ash obtained was analyzed for mineralogical composition using XRD (JDX 3532, JEOL, Japan) and SiO_2 content through an XRF (EDS-7000, Shimadzu, Japan) installed in Central Resource Labs, University of Peshawar. Elemental analysis and surface morphology were investigated using SEM-EDS with JSM-IT100 at the National Centre of Excellence in Geology, University of Peshawar.

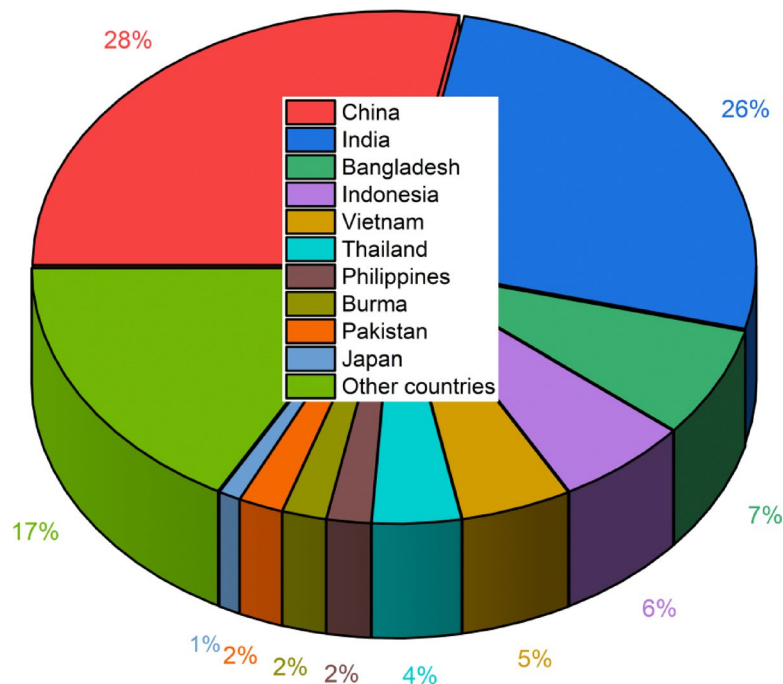


Fig. 2. Percentage of rice production globally for 2023–2024⁵⁸.

Commodity	Unit	% of global production	10-Year average 2013–2024	Production (2022–2023)	Production (2023–2024)	% Change
Rice	Million metric tons	2%	7.74	7.32	9.87	35%

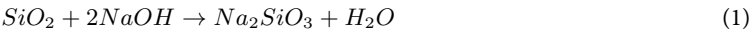
Table 1. Rice production trends in pakistan: 10-year average and recent growth⁵⁹.

Location	Soil type	Major oxide (Wt. %)		
		SiO ₂	Al ₂ O ₃	Fe ₂ O ₃
Buner	Loamy, Shallow	56–75	12–15	2.3–4.2
Charsadda	Loam	50–72	13–19	2.74–6.1
Swat	Loamy, Shallow	43–58	10–18	1–1.76.76

Table 2. Soil properties of the selected regions of Pakistan⁵.

Extraction of Microsilica The RH sample, with maximum silica content obtained after combustion and grinding, was processed into MS using an optimized and economical method. The process for MS extraction was conducted in eleven steps, as mentioned in Fig. 3.

The extraction technique was introduced by Kalapathy⁴⁵ and applied by Lima⁴⁶, khan⁴⁸, and Cavalcante⁴⁹ in their research. In the first phase, RHA was treated with hydrochloric acid (2 M HCl) to eliminate impurities such as Ca, Fe, Na, and K⁴⁸. The mixture was heated and vigorously stirred for 2 h using a magnetic stirrer, then cooled and filtered through Whatman 41 filter paper with a vacuum pump. The treated ash was subsequently subjected to base treatment with 1 M NaOH to solubilize silica, producing sodium silicate as shown in Eq. (1) at stage 5⁴⁶.



The sodium silicate solution was boiled for 2 h, cooled at room temperature, as performed at stages 2 and 3, and filtered to remove remaining solid residues. The filtrate (Na₂SiO₃) was then neutralized (pH 7) using 2 N HCl, forming a gel left for 12 h. Dilute sulfuric acid was added, and the mixture was heated to 90–100 °C until it became acidic (pH 1–2). The reaction that occurred is given in Eq. (2):



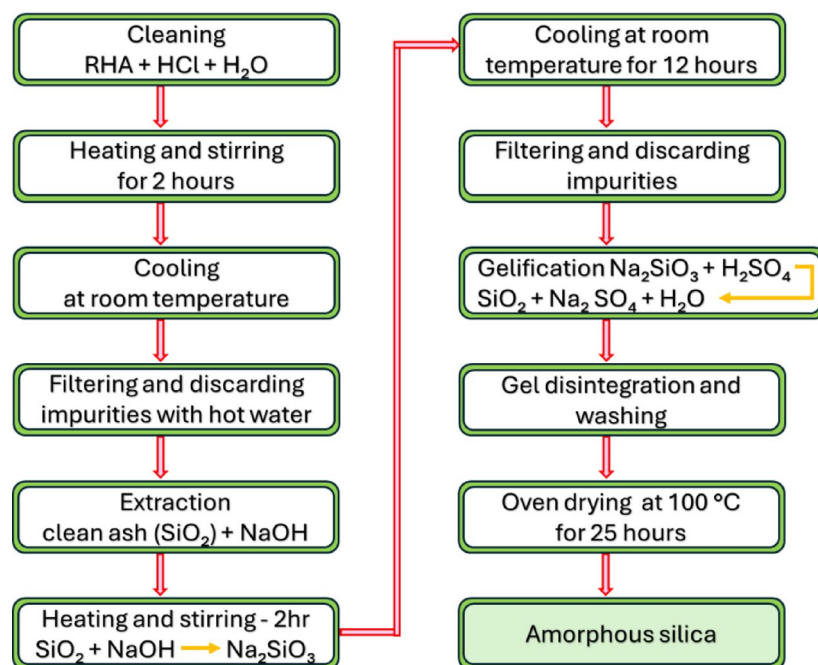


Fig. 3. Flow chart showing the procedure for MS extraction from RHA⁴⁸.

The filtered precipitate was thoroughly washed with distilled water to remove trace impurities. Finally, the residue was placed in an oven at a temperature of 110 °C to obtain amorphous SiO₂ (MS).

The RHA sample, produced through controlled combustion and displaying a distinct grey color, was further analyzed using XRD to evaluate its mineralogical composition. The XRD analysis results show a peak at 20–22° for RHA and EMS samples, as shown in Fig. 4 confirming their amorphous nature⁶⁰. The analysis indicated a typical amorphous structure lacking ordered crystalline arrangements, suggesting a relatively higher disordered silica structure.

XRF analysis was conducted on the RHA sample obtained after burning and on the EMS sample post-extraction to determine the percentage of SiO₂ at CRL, University of Peshawar. The results revealed that MS obtained from chemical processing at PCSIR, Peshawar, contained a high content of SiO₂ (99.470%). Table 3 displays the physical and chemical compositions of OPC, RHA, and EMS.

SEM-EDS analysis was performed to examine the functional groups on the surface of RHA and EMS materials, as displayed in Figs. 5 and 6.

SEM analysis of RHA revealed irregular morphology with visible aggregates, consistent with findings in the literature⁶¹. The fine, irregular particles confirmed the highly amorphous nature of silica, resulting in a high surface area that contributes to enhanced pozzolanic reactivity. According to EDS results, considerable amounts of silica and oxygen, and relatively low calcium suggest that RHA is more effective as a silica contributor rather than a calcium source. The high silica content promotes pozzolanic activity by reacting with Ca(OH)₂ to form CSH gel, improving concrete strength and durability. Similarly, SEM of EMS showed globular structures, facilitating better dispersion in cement⁶². The elemental composition analysis revealed silicon and oxygen as the predominant elements, consistent with the expected SiO₂ content of MS, as confirmed by XRD (Fig. 4). The minor carbon peaks observed in EDS analysis are attributed to standard sample preparation artifacts, specifically from the carbon tape used for mounting⁶³. The reduced particle size and indistinct crystal structures indicate the amorphous nature of the sample⁶⁴. EMS, with high SiO₂ content and minimal impurities, demonstrated excellent pozzolanic properties. Due to its high reactivity, EMS can effectively react with Ca(OH)₂ to form additional CSH gel, significantly enhancing the strength and long-term durability of concrete.

Concrete mix proportions This study evaluated the performance of four concrete mixes, including CM and three ternary mixes incorporating EMS and RHA. In ternary mixtures, high cement replacement ranged from 20% (05% EMS + 15% RHA) to 35% (10% EMS + 25% RHA), and 50% (15% EMS + 35% RHA) was carried out. The EMS replacement levels (5%, 10%, 15%) were selected as per the optimal 05–15% range identified in prior studies for pozzolanic efficiency, while balancing mechanical performance and durability^{46,48,49}. Similarly, the RHA replacement (15%, 25% and 35%) was selected to evaluate the progressive influence of higher replacements on mechanical and microstructural properties⁶⁵. The details of these mixed compositions are presented in Table 4. The primary purpose of adding EMS to ternary mixtures with RHA is to replace the maximum amount of cement without compromising the strength and durability properties of concrete. The total amount of binder for each mixture was 450 kg/m³. Therefore, 157.5 kg/m³ of water was kept the same for all the mixtures. Furthermore, to achieve workability with slump values of 120 ± 30, an additional high-range water-reducing admixture (sikament-520) was utilized at various percentages (wt% of binder).

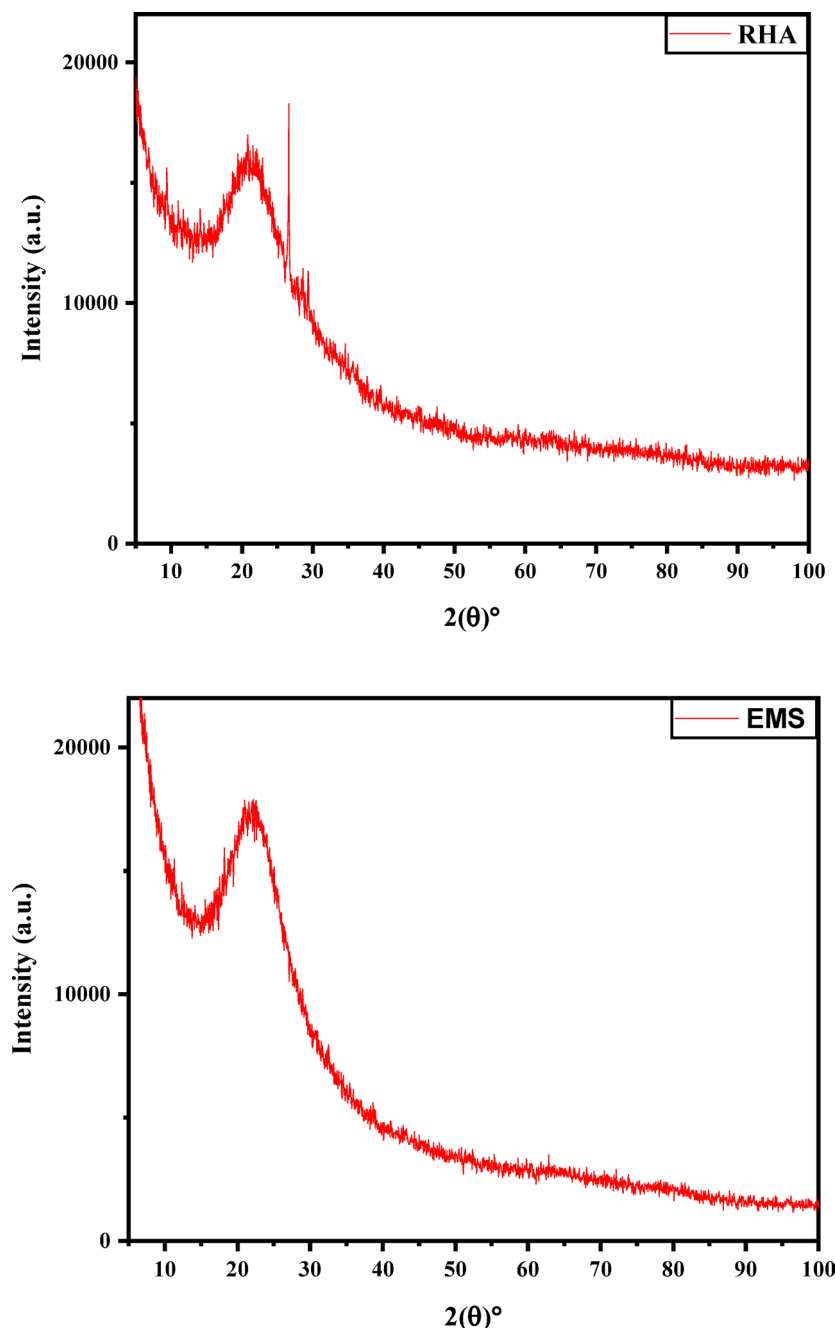


Fig. 4. XRD patterns of RHA and EMS samples.

Testing program

Preparation of specimens

According to ASTM C192⁶⁶ guidelines, concrete ingredients were mixed with the help of a power-driven rotatory mixer. Immediately after achieving the required slump, the freshly mixed concrete was cast into cylindrical molds with a diameter of 100 mm and a height of 200 mm, following ASTM C-39⁶⁷. Samples were prepared for curing periods of 14, 28, and 56 days to evaluate the mechanical strength of concrete. After casting, molds were wrapped with plastic sheets and kept in a controlled environment maintained at 20 ± 1 °C and $60 \pm 5\%$ relative humidity for 24 h. Following this, concrete specimens were demolded and submerged in a water tank for the specified curing ages of 14, 28, and 56 days. Before testing, the top and bottom surfaces of the specimens were leveled with the help of an end surface grinder to ensure uniformity.

Compressive and splitting tensile strength

Compressive and splitting tensile tests were performed on the concrete cylinder specimens at the curing ages of 14, 28, and 56 days. The test was carried out using a 200-ton Universal Testing Machine (UTM, SHIMADZU,

	OPC	RHA	EMS
Fineness (m ² /g)	-	0.360	36.58
Specific gravity (Kg/m ³)	3.15	2.83	2.15
Particle size (nm)	-	1273	487.5
Chemical composition (oxides, % by weight)			
SiO ₂	21	91.028	99.470
CaO	61.7	2.225	0.252
Al ₂ O ₃	5.04	-	-
Fe ₂ O ₃	3.24	2.134	0.064
P ₂ O ₅	-	1.049	-
MgO	2.56	-	-
SO ₃	1.51	0.472	0.115
K ₂ O	0.62	2.789	0.056
Na ₂ O	0.13	-	-
TiO ₂	-	0.066	0.007
MnO	-	0.137	0.006
CuO	-	0.016	0.008
Free lime	0.68	-	-
IR	0.54	-	-
LOI	1.83	-	-

Table 3. Physical and chemical compositions (% by weight) of raw materials.

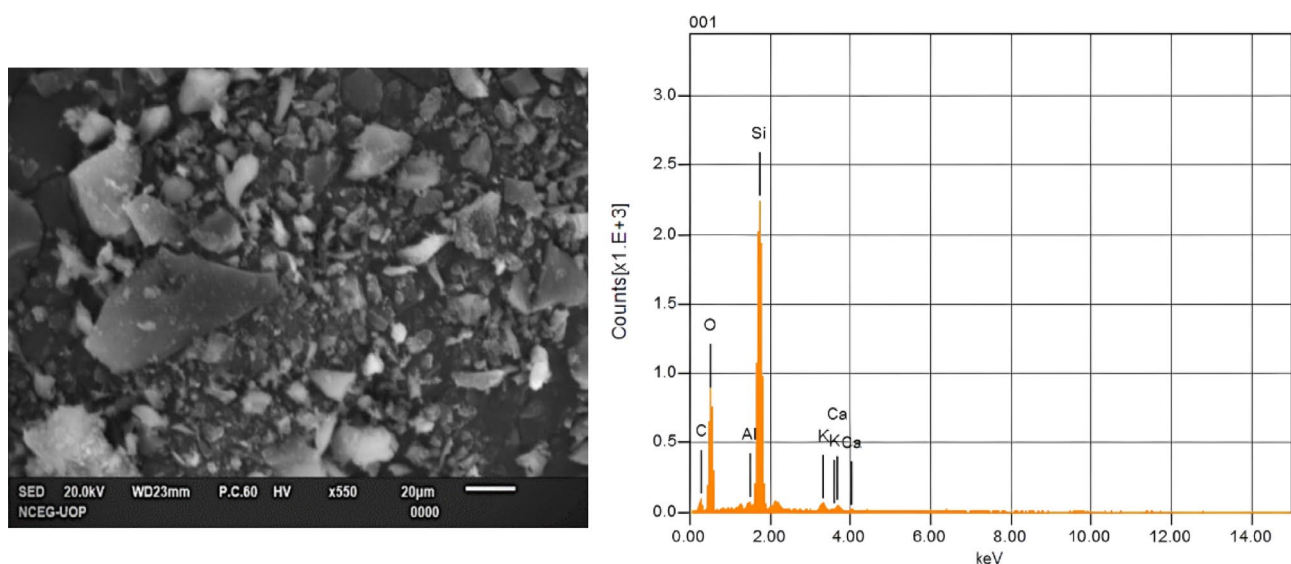


Fig. 5. SEM micrograph with EDS for RHA sample.

Japan) at the Sarhad University of Science and Information Technology. The strength results were determined by averaging the values obtained from the tested samples. The compressive strength was determined using the formula given in Eq. (3), as per ASTM C39 standard⁶⁷, with load applied at a constant rate of 0.2 MPa/s. On the other hand, the splitting tensile strength was determined using Eq. (4), following ASTM C-496⁶⁸, with load applied at a constant rate of 1 MPa/min. Figure 7 illustrates the experimental setup employed to determine the compressive and splitting tensile strengths of the concrete samples.

$$f'_c = P/A \quad (3)$$

Where P represents the maximum load (N), A denotes the cross-sectional area (mm²), f'_c is the compressive strength (MPa).

$$T = 2P/\pi ld \quad (4)$$

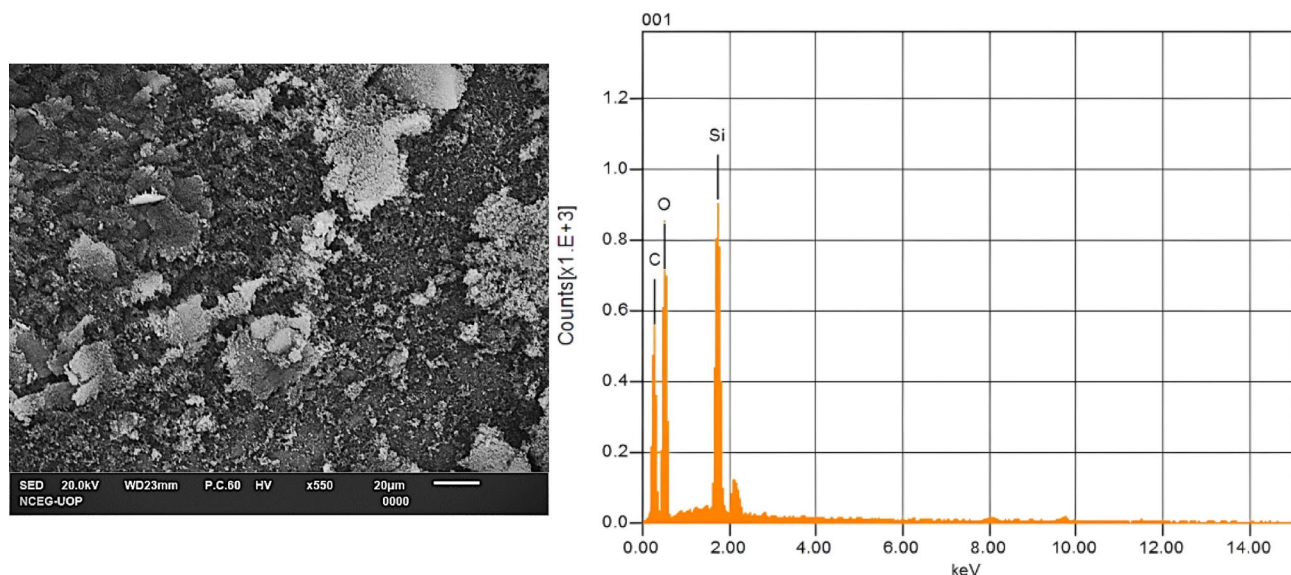


Fig. 6. SEM micrograph with EDS for EMS sample.

Mix ID	Water, w (kg/m ³)	Binder, bi (kg/m ³)			Aggregates, ag (kg/m ³)			SP (% of bi)	Slump (mm)
		OPC	EMS	RHA	FA (s)	CA 20 mm	CA 10 mm		
Control mix (CM)	157.5	450	-	-	700	682	418	1.0	120 ± 30
Concrete containing 05% EMS and 15% RHA (E05R15)		360	22.5	67.5				1.2	
Concrete containing 10% EMS and 25% RHA (E10R25)		292.5	45	112.5				1.3	
Concrete containing 15% EMS and 35% RHA (E15R35)		225	67.5	157.5				1.4	

Table 4. Mix details of concrete samples for mechanical tests ($w/bi = 0.35$; $ag/bi = 4.0$; $s/ag = 0.38$).



Fig. 7. Test setup for measuring compressive and splitting tensile strength of concrete samples.

Where d denotes the diameter of the specimen (mm), l is the length of the specimen (mm), and T is the splitting tensile strength (MPa).

Water absorption and apparent porosity (AP)

In addition to mechanical testing, WA and AP tests were conducted on the hardened concrete samples per ASTM C948⁶⁹. These tests were performed to assess the concrete's water absorption capacity, providing valuable insight into its durability, including porosity and permeability characteristics. Cylindrical specimens (100 mm

diameter, 50 mm thickness) were prepared from samples cured for 56 days. These specimens were immersed in water at 21 °C, and their weights were observed at 24 h intervals until a stabilized saturated surface dry (SSD) condition was attained. The SSD was considered to be achieved when the weight difference between two consecutive measurements was less than 0.5%. The most recent weight recorded is denoted as 'B'. Subsequently, the samples were suspended in water, and their weight was determined to be the nearest 0.01 g, recorded as 'A'. The specimens were dried in an oven at 100 ± 5 °C, and their weights were noted at 24 h intervals. Once the weight loss was observed to be less than 0.5% of the previous reading, the specimens were allowed to cool in a vacuum desiccator, maintaining room temperature. The recorded weight at this stage is designated as 'C'. Using Eqs. (5) and (6), WA and AP values were then calculated.

$$WA(\%) = (B - C)/C \times 100 \quad (5)$$

$$AP(\%) = (B - C)/(B - A) \times 100 \quad (6)$$

Concrete permeability

Water permeability test was performed on the hardened concrete specimens to investigate water penetration depth at the National University of Sciences and Technology, Islamabad, Pakistan, following BS EN 12390-8⁷⁰. Three cylindrical samples (300 mm diameter, 150 mm height) from each mixture, water-cured for 56 days, were evaluated under a water pressure of 500 ± 50 kPa on the bottom surface. Observations were made periodically on the unpressurized surface to monitor any changes in appearance. After 72 ± 2 h, the specimens were split using a Universal Testing Machine to investigate the depth of water penetration. The average penetration depth was calculated from the measurements taken from three cylindrical samples; all cast with the same concrete batch and evaluated simultaneously. Water penetration under exterior hydraulic pressure in partially saturated media, such as concrete, is linked to sorptivity and unsteady flow. Thus, the method quantifying water transport under a pressure gradient can be evaluated under a non-stationary regime⁷¹. The primary assumption is that the water penetration test is performed only when steady-state conditions are reached⁷². However, the studies presented indicate that applied pressure and measurement duration are adequate to achieve results corresponding to a fully saturated state, even for concrete with low permeability. Furthermore, the high density of tested concrete samples with minimal penetration depth relative to the specimen supports the uniaxial water penetration assumption. Consequently, this method enables us to calculate the coefficient of permeability.

The permeability coefficient can be determined using Eq. (7), derived from the modified Valenta equation referenced by Ramachandran⁷¹ and Skutnik⁷².

$$k = \alpha d^2 / (2Ht) \quad (7)$$

whereas 'k' is the permeability coefficient (m/s), 'd' the water penetration depth, 'α' is the permeable porosity, 't' test time (s), and 'H' is the hydrostatic head causing percolation of water (m).

Microstructure investigation

Preparation of cement pastes for microstructure investigation CM specimen and three ternary paste specimens with different proportions of EMS and RHA (E05R15, E10R25, and E15R35) as partial cement replacements (by weight) were prepared. The mixer was thoroughly blended using a Hobart mixer. After the proper mixing, the pastes were poured into circular plastic molds 20 mm in diameter and 50 mm in height. Subsequently, the molds were sealed and kept for 28 days to cure, maintaining room temperature. After curing, the test specimens were demolded and dried using an ion exchange method to ensure the stoppage of the hydration process. The acquired powdered and thinly sliced specimens were cleaned in isopropanol for 15 min, then oven-dried at 40 °C for 30 min. The specimens obtained were held in polythene plastic bags till testing.

X-ray diffraction The mineralogical investigation of well-ground cementitious samples was conducted using a specially designed sample handler for analysis of an X-ray powder diffraction machine using (XRD, JDX 3532, JEOL, Japan, with SX-Rays-Cu Ka ($I = 1.5418$ Å)). The XRD data were obtained in the range of $2\theta = 0^\circ$ to 80° , installed in the Central Resource Labs, University of Peshawar. After 28 days, the cured samples were taken out of their molds and crushed into smaller fractions, which were treated with ethanol to stop further hydration⁷³. Before analysis, the smaller fraction was further oven-dried and ground to a particle size lower than $70 \mu\text{m}$ ⁷⁴.

SEM-EDS analysis The microstructural behavior of cement pastes was studied using the SEM-EDS technique on thick slices prepared from the hardened specimens with JSM-IT100 at NCEG, University of Peshawar. As discussed above, the hardened paste slices were dried through the solvent exchange method using isopropanol. The microstructural and compositional transformations of the tested specimens were examined.

Fourier transformed infrared (FTIR) FTIR analysis was performed to characterize the different functional groups present in the cement paste samples having various percentages of EMS and RHA using an FTIR Spectrometer (Model Cary 630, Agilent Technologies, USA) in addition to the microstructural investigation of samples. The infrared spectrum was recorded between (400 cm^{-1} and 4000 cm^{-1}) when the dry powder specimens were directly exposed to an infrared light source.

N₂ adsorption Besides XRD and SEM analysis, N₂ adsorption tests were conducted on powdered samples (approximately 0.3 g) collected from the control and ternary-based mixes after 28 days of casting. An N₂ adsorption analysis was performed to find the induced pore volume and BET surface area using (NOVA2200e, Quanta

	Compressive strength (MPa)			Split tensile strength (MPa)		
	Age (days)					
Mix designation	14	28	56	14	28	56
CM	24.45 (± 0.98)	30.67 (± 0.77)	35.64 (± 1.08)	2.29 (± 0.11)	2.68 (± 0.16)	3.12 (± 0.23)
E05R15	28.23 (± 1.04)	34.22 (± 1.09)	42.08 (± 1.76)	2.73 (± 0.12)	3.45 (± 0.12)	4.32 (± 0.04)
E10R25	22.55 (± 1.38)	32.51 (± 1.47)	39.14 (± 0.73)	2.39 (± 0.09)	3.18 (± 0.13)	3.82 (± 0.20)
E15R35	20.88 (± 1.16)	26.82 (± 0.52)	33.31 (± 1.18)	2.01 (± 0.19)	2.52 (± 0.06)	3.04 (± 0.14)

Table 5. Compressive and split tensile strength results of CM and ternary mixtures.

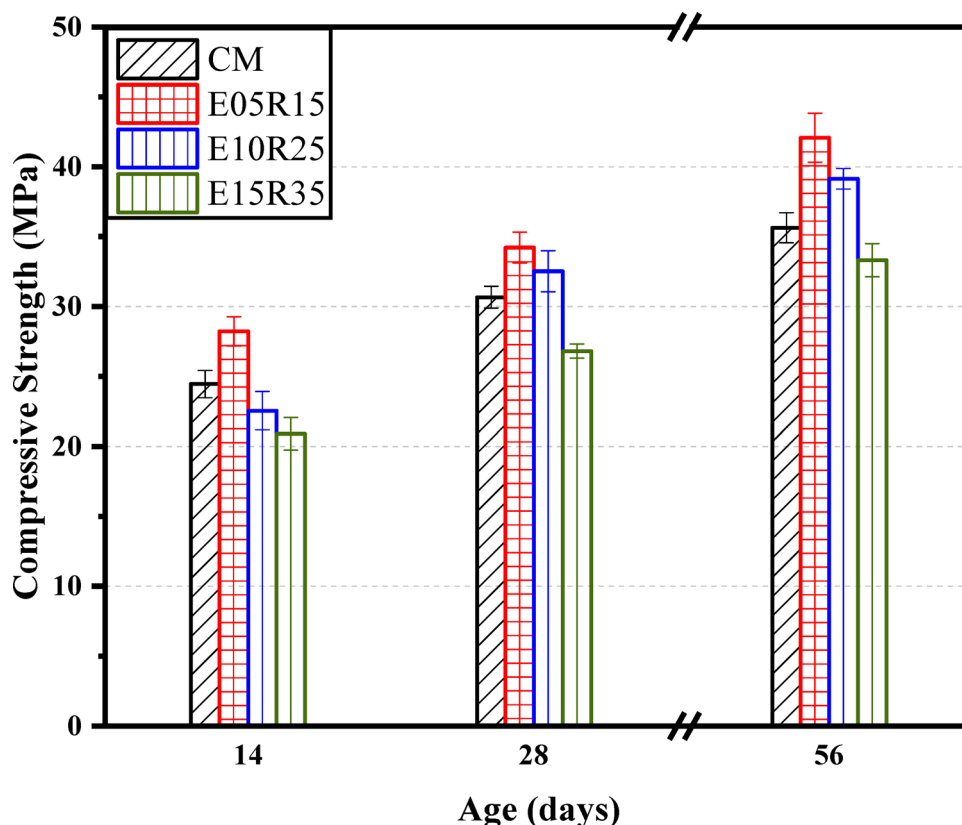


Fig. 8. Comparison of compressive strength results for and ternary mixes.

chrome, USA) analyzer, maintaining the temperature of 273 K, at the Central Research Laboratory, University of Peshawar. During the procedure, the powdered samples were initially degassed to eliminate any airborne contaminants. Similarly, the test was performed while maintaining controlled pressure and ambient temperature.

Results and discussion

Influence of EMS and RHA on the compressive strength with aging

The compressive and splitting tensile strength results of all tested concrete samples at 14, 28, and 56 days are displayed in Table 5. The results are presented as an average for identical samples, with a standard deviation ($\pm$) at each curing age. Figure 8 displays the compressive strength results for CM and ternary-based samples incorporating various percentages of EMS and RHA (CM, E05R15, E10R25, and E15R35). The study aims to assess the strength performance of these ternary-based mixtures against CM and identify the optimum percentage replacement of cement without compromising on the strength. Before testing, the samples were cured under standard conditions of 20 °C until the designated testing age.

The test results reveal that the EMS and RHA mixes (E05R15 and E10R25) displayed a noticeably better compressive strength compared to the CM at all curing ages, except for a slight decrease for E10R25 at 14 days of curing. In contrast, the E15R35 sample exhibited lower compressive strength than CM at all ages, with the lowest strength recorded for samples with high percentages of replacement by EMS and RHA (E15R35). This was attributed to the slower pozzolanic reaction, adversely affecting the hydration process and making the concrete more porous, leading to a reduction in strength. While E10R25 experienced a slight decrease in strength at

14 days compared to CM, an improvement in strength at later ages, particularly at 56 days, is attributed to the delayed pozzolanic activity of RHA.

At 14 and 28 days, the concrete with 5% EMS and 15% RHA substitution exhibited a compressive strength that was 15.40% and 11.56% higher than the CM, respectively, as shown in Table 5. In contrast, the strength of the E10R25 sample at 14 days was 7.77% lower than that of the CM, while it increased by 5.98% at 28 days. For the E15R35 sample, the strength was reduced by 14.57% at 14 days and 12.55% at 28 days compared to CM. To achieve high sustainability, a significant portion of cement was replaced by RHA, combined with different percentages of EMS to compensate for the strength reduction caused by higher RHA replacement. This remarkable increase in strength for E05R15 and E10R25 mixes is mainly due to the development of denser CSH gel, as validated in SEM-EDS analysis. According to Previous literature, researchers concluded that incorporating reactive amorphous SiO₂ with pozzolanic properties and high fineness in SF significantly improves cement mortar and concrete strength⁷⁵. Lima⁴⁶ and Cavalcante⁴⁹ also noticed an improvement in cement paste compressive strength at 5% and 16% cement substitution by silica obtained from RHA. The results revealed that the early strength increase is attributed to an improved hydration process and better voids filling ability of EMS, in addition to later-age improved pozzolanic reactivity of RHA. Furthermore, compressive strength was reduced by 50% by replacing cement with all mixes. This decrease in compressive strength is attributed to the high percentage replacement of cement, which causes increased agglomeration due to the overabundance of silica, which lowers the hydration process and forms weak products⁷⁶. Furthermore, the E10R25 sample shows only a 5.98% strength increase; its practical viability is justified by meeting industry strength tolerances, superior durability (lower porosity compared to CM), and optimized microstructure (denser CSH gel). The 35% cement replacement (25% RHA + 10% EMS) maintains workability while fully utilizing agro-industrial byproducts, making it ideal for applications where durability outweighs marginal strength gains (e.g., marine environments).

Influence of EMS and RHA on the split tensile strength with aging

Figure 9 compares the splitting tensile strength of ternary concrete mixes with varying percentages of EMS and RHA (E05R15, E10R25, and E15R35) with that of CM with aging. Ternary mixes with lower replacement levels (E05R15 and E10R25) exhibited higher splitting tensile strength compared to CM at all curing ages. In contrast, the mix with a high replacement level of 50% (E15R35) shows reduced splitting tensile strength as compared to CM across all curing ages. The results aligned with the compressive strength trends, except for the E10R25 sample, which demonstrated higher splitting tensile strength than CM at all curing ages despite lower compressive strength at 14 days.

The splitting tensile strength values for the E05R15 and E10R25 at 14 days were 19.46% and 4.63% higher than CM, respectively, as shown in Table 5. A similar strength increase was observed at 28 and 56 days of curing.

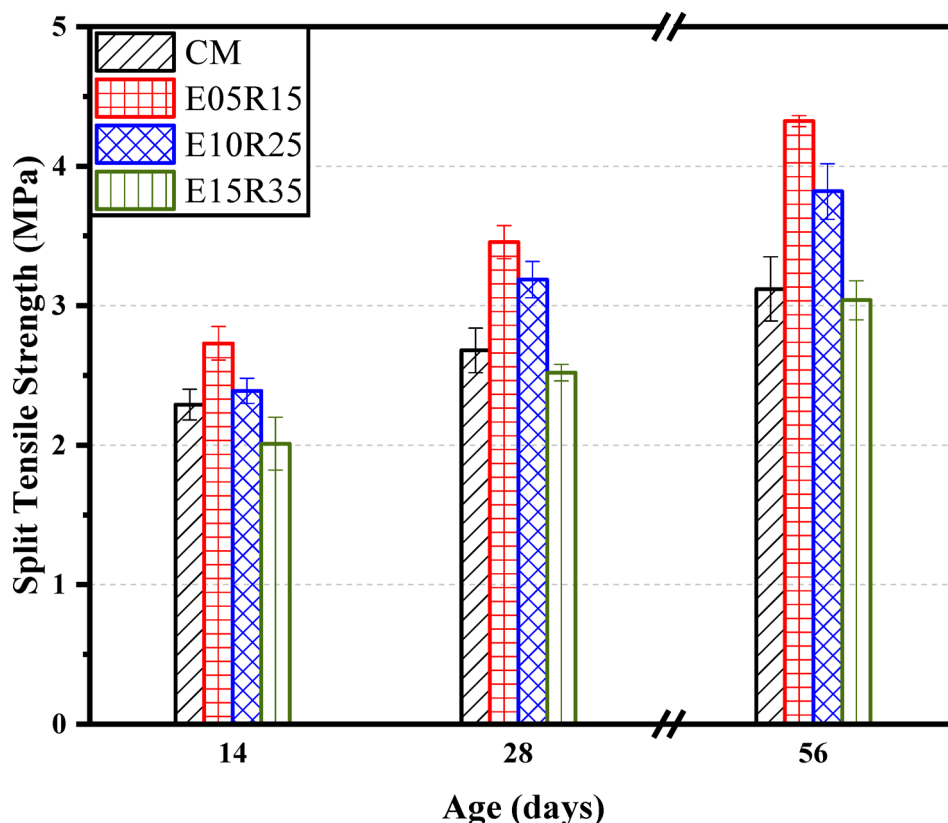


Fig. 9. Comparison of tensile strength results for CM and ternary mixes.

However, the E15R35 sample exhibited a 12.22% lower strength compared to CM at 14 days, continuing the same decreasing trend at 28 and 56 days. The significant increase in splitting tensile strength of ternary mixes at early ages suggests that EMS enhances early hydration reactions at a low percentage of replacement. EMS, having a pozzolanic nature, reacts with calcium hydroxide (formed during cement hydration), resulting in additional CSH gel, which contributes to increased strength, a denser and more refined microstructure, and improved interlocking between cement and aggregates⁴⁸. In contrast, the ternary mix E15R35 with high percentage replacement exhibited lower splitting tensile strength compared to CM at all curing ages. This reduction in split tensile strength is due to the high percentage of cement replacement that adversely affects the hydration reaction due to the overabundance of silica, resulting in a more porous structure with cracks. The improved splitting tensile strength for E05R15 and E10R25 samples at 28 and 56 days of age might be attributed to the long-term strength development driven by RHA's pozzolanic activity, which continues to produce additional CSH gel⁷⁷. There is a strong connection between the compressive and splitting tensile strength of concrete. Various standards and researchers have proposed empirical equations to describe this relationship, which are summarized in Table 6. Figure 10 presents a comparison of experimental data with predictive models from various codes and researchers. The results revealed that the proposed equation by Zain et al.⁷⁸ provides the most accurate prediction, achieving a coefficient of determination (R^2) of 0.824. This is closely followed by CEB-FIB⁷⁹, which achieves an R^2 value of 0.81.

Comparison of results obtained for the water absorption and apparent porosity of control and ternary mixes

In addition to validating the strength properties of the ternary concrete mixtures under investigation, other key characteristics, such as WA and AP, were also assessed and compared with the CM. WA and AP are commonly used as indirect indicators for evaluating the durability characteristics of hardened concrete. The influence of ternary mixtures having different percentages of EMS (5%, 10%, and 15%) and RHA (15%, 25%, and 35%) on WA and AP is displayed in Figs. 11 and 12, respectively. The results revealed that the low cement replacement by EMS and RHA by 20% (E05R15) exhibited the lowest WA and AP values compared to the CM and higher replacement levels (E10R25 and E15R35), as shown in Figs. 11 and 12. The decrease in WA of the E05R15 and E10R25 mixes can be attributed to the low value of AP (Fig. 12), which is likely due to the fine particles of EMS and RHA, contributing to significant pore refinement in the concrete. The incorporation of RHA up to 20% cement replacement leads to a significant reduction in WA compared to CM⁸⁴. In contrast, the higher replacement level of EMS and RHA (E15R35) resulted in increased WA and AP values compared to CM. The higher WA and AP values observed with increased cement replacement by EMS and RHA up to 50% might be attributed to the slower pozzolanic reaction rate, adversely affecting the hydration process and resulting in a more porous microstructure. The findings also confirm that the compressive and split tensile strengths of concrete samples containing EMS and RHA are higher than those of CM, except for concrete samples with 15% EMS and 35% RHA (E15R35), whose strengths were lower than CM. Higher RHA content can reduce mechanical strength and increase water absorption and porosity due to unreacted silica, which limits its participation in hydration and leads to weaker hydration products⁸⁵. Moreover, high dosages of EMS tend to form agglomerates due to their ultrafine particle size, which hinders proper hydration and results in weaker products, ultimately reducing strength and durability^{48,86}. The above results revealed that cement replacement up to 10% by EMS and 25% by RHA may be adopted for future applications, as it achieves satisfactory strength results. Additionally, replacing cement with EMS and RHA at high levels, like 50%, is not recommended, as it leads to reduced strengths at all ages compared to CM.

Comparison of results obtained for the water permeability of the control mix and ternary mixes

Water permeability is a critical property influencing the long-term durability of concrete. Higher values of water permeability result in increased water penetration, which elevates the risk of deterioration, especially when concrete is exposed to harsh environments, such as seawater or industrial water. The increased water penetration, in turn, compromises the durability of concrete structures in the long term. The results of water penetration and the permeability coefficient derived from penetration are illustrated in Table 7. The test results revealed that incorporating EMS and RHA at low percentage replacements (E05R15 and E10R25) significantly reduced the water penetration compared to CM and the high percentage replacement mixture by 50% (E15R35). Specifically, water penetration decreased by 32.67% for E05R15 and 11.63% for E10R25

Model code	Model equation	Values of	
		a	b
ACI 318–2014 ⁸⁰	$f_{sp} = a (f'c)^b$	0.56	0.50
ACI 363R-92 ⁸¹		0.59	0.50
CEB-FIP 1990 ⁷⁹		0.301	0.67
JSCE-2012 ⁸²		0.23	0.667
Tomosawa et al. ⁸³		0.23	0.667
Zain et al. ⁷⁸	$f_{sp} = f'c/[a (f'c) + b]$	0.10	7.11

Table 6. Comparison between compressive and splitting tensile strength of concrete with existing prediction equations.

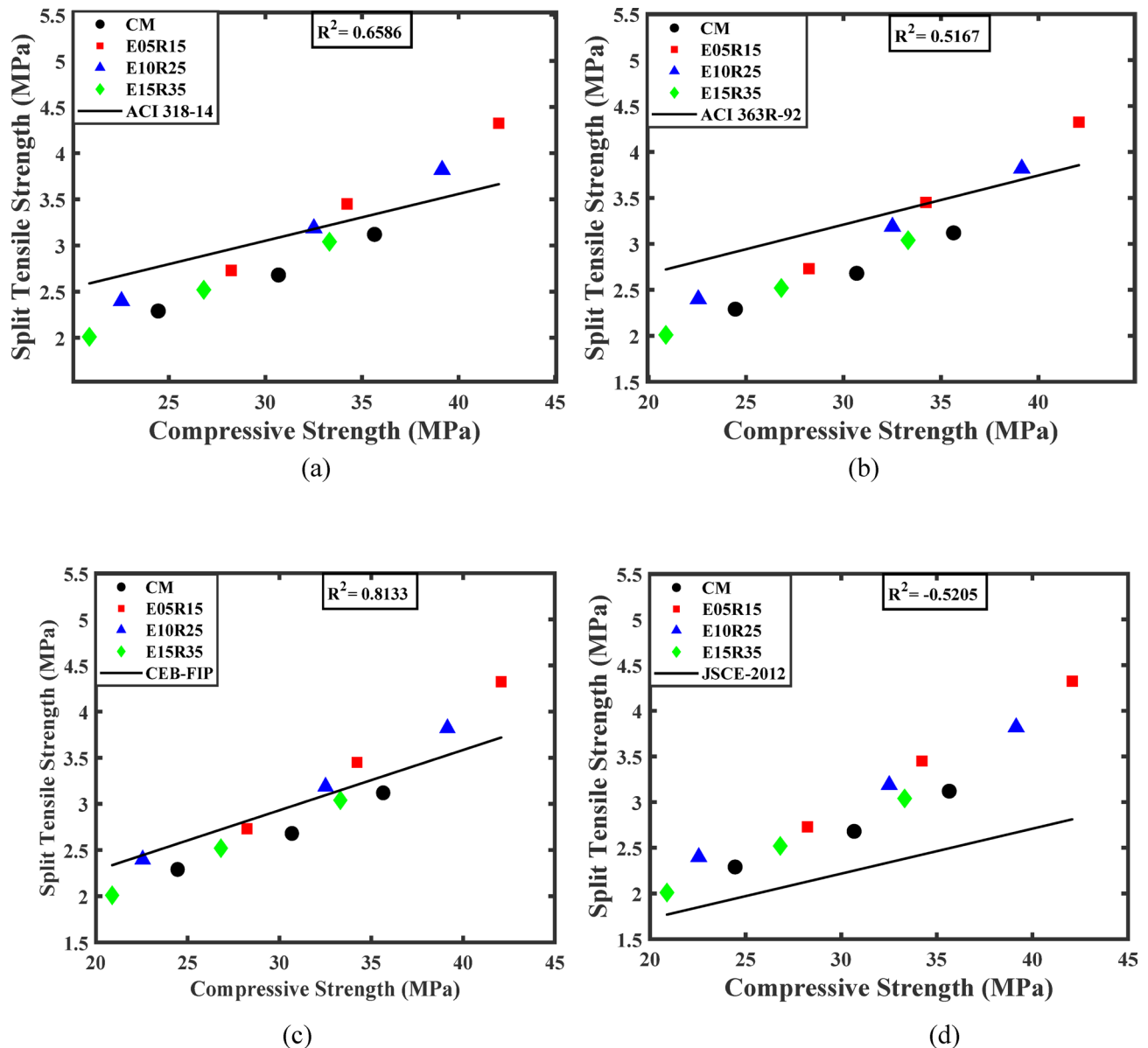


Fig. 10. Correlation between compressive and tensile strength of concrete: experimental results with different proposed model equations.

compared to CM. E10R25 showed a significant reduction in water penetration as compared to the E05R15. However, its consistently higher compressive and tensile strength across all curing periods ensures superior structural performance for demanding applications. The observed water penetration reduction results from the synergistic filler effects of EMS and RHA particles, which enhance matrix density and reduce porosity relative to the control mixture. For durability-critical applications, the combination of improved mechanical properties and moderate permeability reduction makes E10R25 the optimal choice, as structural integrity often outweighs permeability considerations alone. The addition of pozzolans such as RHA and SF decreases the pore size and makes the concrete less permeable⁸⁷. In contrast, the higher percentage replacement sample, E15R35, exhibited increased water penetration by 29.70% compared to CM. The increase in water penetration by adding high percentages of EMS and RHA is attributed to poor dispersion, causing agglomeration problems, adversely affecting the hydration process, and ultimately resulting in weak products. Inadequate dispersion of silica promotes agglomeration, impairing hydration, liquid penetration, and stability, particularly when combined with a high percentage of RHA⁸⁸. Wu et al.⁸⁹ reveal that poor dispersion of SF results in particle agglomeration, leading to weaker microstructure development and reduced stability compared to well-dispersed samples. The E05R15 ternary sample reflected superior performance in resisting water penetration, followed by E10R25, CM, and E15R35. Furthermore, all the values obtained for concrete mixtures are in the range of 10^{-9} to 10^{-10} mm/s and are desirable for concrete used in structures exposed to severe environments, such as marine and industrial conditions.

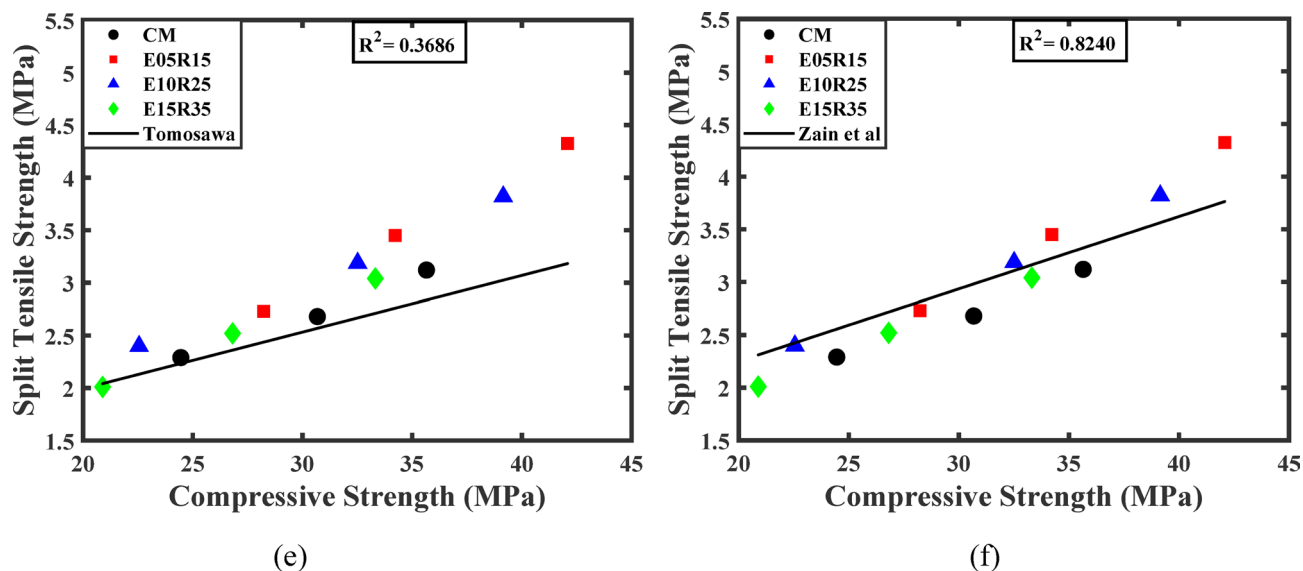


Fig. 10. (continued)

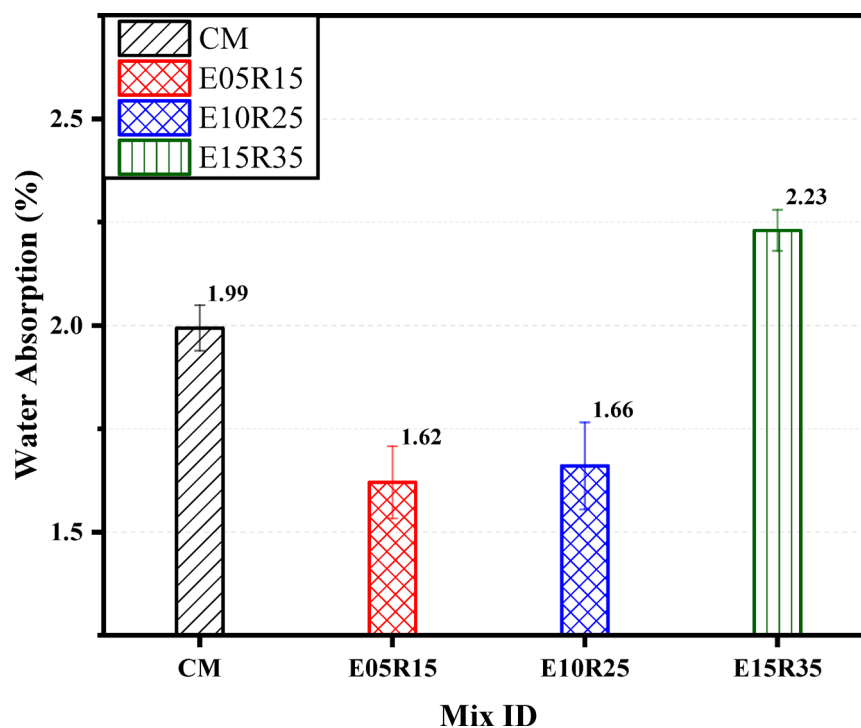


Fig. 11. Comparison of water absorption results for CM and ternary mixes.

Adding RHA and EMS within specific limits decreased the water permeability of concrete mixtures, primarily due to the formation of a dense and less porous microstructure. The findings demonstrate that the permeability of concrete mixtures has a direct impact on their strength. Higher permeability levels suggest the presence of a greater number of pores within the concrete specimens, which reduces their strength and allows water to infiltrate. This infiltration can lead to issues such as corrosion, cracking, and overall deterioration of the concrete. In this current study, similar trends of reduction in permeability and increase in strengths are observed, along with increasing percentage replacement by EMS and RHA up to certain limits, as presented in Fig. 13. The increase in strength with the decrease in the water permeability is attributed to the formation of a less porous and dense microstructure⁹⁰.

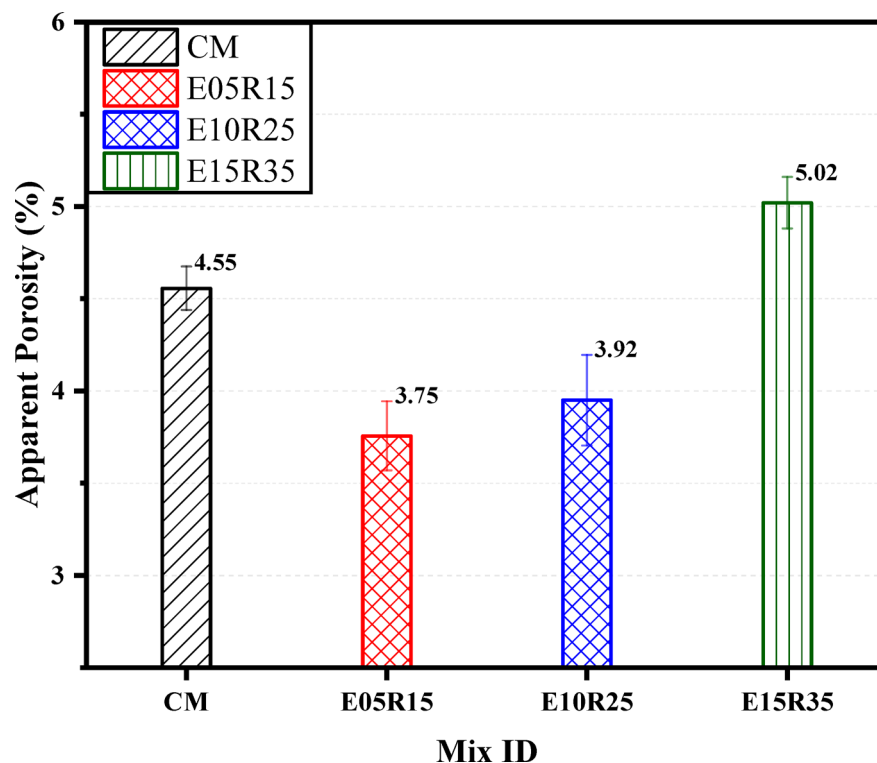


Fig. 12. Comparison of apparent porosity results for CM and ternary mixes.

Mix designation	Water penetration depth (mm)	Permeability coefficient (mm/s)
CM	40.4	2.77×10^{-9}
E05R15	27.2	8.98×10^{-10}
E10R25	35.7	1.71×10^{-9}
E15R35	52.7	5.29×10^{-9}

Table 7. Comparison of water penetration depth and permeability coefficient results for control and ternary mixes.

X-ray diffraction analysis

The XRD analysis of the CM and specimens containing EMS and RHA is shown in Fig. 14. The XRD data contains mainly portlandite, ettringite, silicates, and quartz. It has been noted that the intensity of the portlandite ($\text{Ca}(\text{OH})_2$) has been reduced with the increase in the amount of MS⁷⁴. However, this decrease in the portlandite for EMS and RHA replacement levels is due to the formation of a higher amount of additional CSH gel, formed by the reaction of MS with $\text{Ca}(\text{OH})_2$. This formation of CSH gel is more dominant in the E05R15 and E10R25 samples, illustrated by its higher mechanical properties. Cavalcante⁴⁹ also provided evidence that the amount of portlandite was reduced with the increasing substitution of cement by extracted silica through performing thermal analysis. The increased portlandite consumption with the incorporation of extracted silica and RHA indicates its significant pozzolanic reactivity. The quartz peak observed at high replacement levels of EMS and RHA (E15R35) is attributed to unreacted silica, resulting from the excessive replacement of cement with RHA and high dosages of EMS, as confirmed by SEM analysis. The results obtained from the XRD graph align with SEM-EDS and FTIR data, providing evidence of high consumption of portlandite, resulting in the subsequent formation of denser CSH gel at lower percentages of EMS and RHA substitution levels.

SEM-EDS analysis

The results of SEM micrographs for the CM and the ternary-based mixes are demonstrated in Fig. 15. The effect of EMS and RHA as a substitution for cement with different percentages was observed. The microscopic study illustrates that E05R15 and E10R25 substitutions exhibited a denser and more compacted matrix than the CM. These results may be related to EMS's high reactivity, which produces high-density hydrates and a well-refined microstructure. This improvement in strength and microstructure is attributed to the formation of additional CSH gel, leading to a more compact and durable geopolymer structure⁹¹. However, the SEM micrographs also show un-hydrated cement and agglomeration of MS grains in addition to some cracks at high percentage

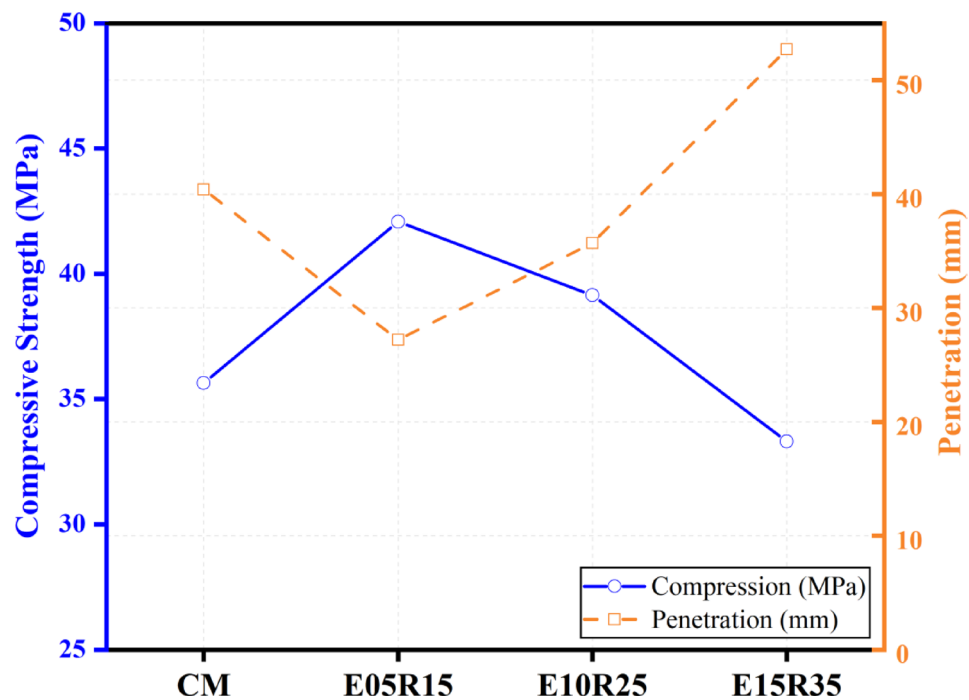


Fig. 13. Relationship between compressive strength and permeability at 56 days of age.

replacement by EMS and RHA (E15R35), resulting in a less dense and porous microstructure, as shown in Fig. 15. These accumulated particles may be associated with the over-saturation and poor dispersion of EMS and RHA in the cement paste. The overabundance of reactive silica could cause reduced hydration reaction, and it may be responsible for the lower consumption of anhydrous cement, forming an agglomeration of MS and large voids, ultimately resulting in decreased strength and durability performance⁹².

SEM micrograph and corresponding EDS elemental analysis for different mixes were additionally performed, as shown in Fig. 16. The purpose was to calculate Ca/Si ratios for the cement paste matrix, considered a crucial parameter for analyzing the performance of mineral admixtures utilized in cement for designing high-strength and sustainable concrete with reduced carbon emissions⁹³. From the results, the value of Ca for the sample E15R35 was lower than all the respective samples. The reduction in Ca content is mainly due to the consumption of more Ca for forming CSH gel. The Ca/Si value for the CM is high compared to all other samples. With the increase in EMS and RHA percentages, the Ca/Si value decreases, which indicates that additional silica from EMS and RHA reacts with CH to produce more CSH gel. E10R25 sample strength retention despite higher porosity is explained by its moderately lower Ca/Si ratio, which correlates with the SEM-observed denser microstructure. This suggests preferential formation of refined gel pores (rather than capillary voids) in the polymerized CSH matrix (lower Ca/Si), creating a cohesive microstructure that maintains strength while exhibiting greater total porosity.

FTIR spectroscopy

The FTIR spectroscopy compares the results between the CM and mixes containing EMS and RHA substitutes with different percentages, and all samples exhibit IR bands at the exact location, while their intensities vary, as demonstrated in Fig. 17. These variations in the intensities primarily result in the production of CH and CSH, major hydration products. In the range between 450 cm^{-1} and 1100 cm^{-1} , all samples exhibit a complicated collection of bands demonstrating symmetric and asymmetric stretching of the Si-O bands. In particular, the prominent peaks at 900–1100 cm^{-1} are linked with Si-O band vibrations in the CSH phase⁹⁴. The peak shifts in these ranges indicate silica polymerization. The absorption peak recorded at 954 cm^{-1} in the control sample indicates the valence fluctuations of the Si-O band in a tetrahedral structure. This peak is associated with the linkage of the oxygen atom [SiO_4] to the central Si-O atom, which plays a role in CSH formation. Peak shifts for all ternary mixes (EMS and RHA) substitutions from 954 to 961 cm^{-1} , respectively. As a result of the development of additional SiO_2 gel, these shifts exhibit prominent levels of silica polymerization with the incorporation of EMS and RHA. The FTIR spectroscopy results align with the EDS results, indicating that E05R15 and E10R25 samples formed denser CSH-phases having a lower Ca/Si ratio than the control sample. The increase in compressive strength for E05R15 and E10R25 mixes can be linked to the production of additional CSH gels. The observed Si-O peak shift (954 to 961 cm^{-1}) in E15R35 without enhanced polymerization reflects silica agglomeration, validated by SEM analysis, leading to compromised silica connectivity, despite higher silica content. Additionally, the intensity peaks were recorded at 711 cm^{-1} , 872 cm^{-1} , and 1420 cm^{-1} as a result of calcite formation during carbonation⁹⁵.

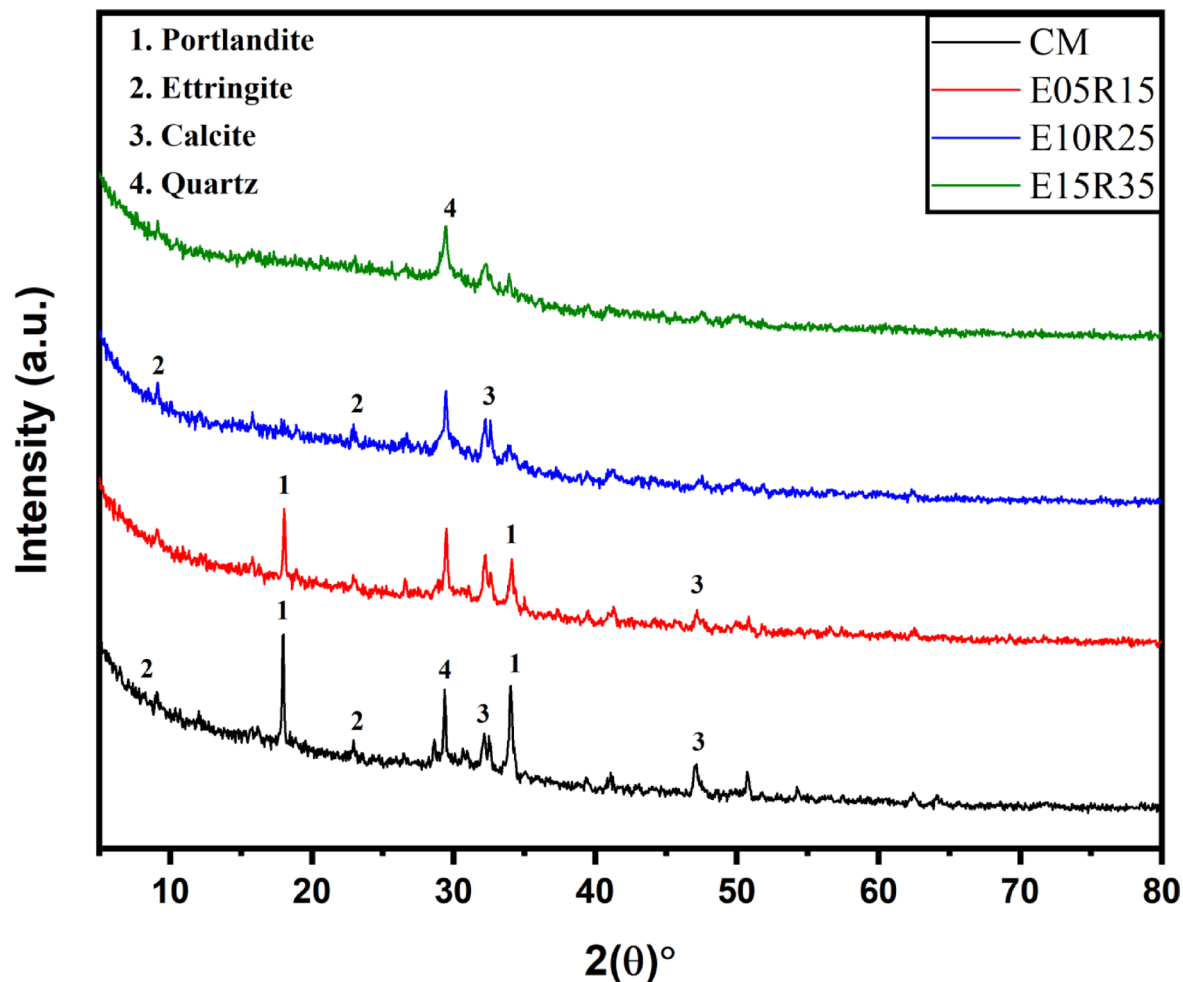


Fig. 14. XRD patterns for control paste cement and ternary pastes with different percentages after 28 days.

In CM sample, the intensity of the FTIR peak at 3641 cm^{-1} confirms the availability of OH groups, indicating the occurrence of the portlandite phase, which has low strength due to a lamellar structure. Notably, the E05R15 sample shows a modest Portlandite peak, while the E10R25 and E15R35 substituted samples show considerably smaller or completely disappeared peaks, indicating higher pozzolanic reactivity of EMS and RHA than CM. The observed reduction in the peak intensities suggests the consumption of CH, leading to the development of additional CSH gels in ternary-based mixtures. The SEM-EDS analysis revealed similar trends, showing an increase in the formation of high-density CSH gels and a decrease in $\text{Ca}(\text{OH})_2$ content with the incorporation of EMS and RHA, particularly in the E05R15 and E10R25 paste samples.

N₂ adsorption

Table 8 lists the BET (Brunauer–Emmett–Teller) surface area and the BJH (Barrett–Joyner–Halenda) pore volume measurements, determined using nitrogen adsorption analysis after 28 days of curing. The findings showed that the ternary mix containing 20% cement substitution (E05R15) exhibited the lowest nitrogen volume intrusion ($0.002\text{ cm}^3/\text{g}$), followed by E10R25, E15R35, and the CM (Table 8). This behavior can likely be explained by incorporating highly reactive EMS, resulting in improved pozzolanic activity and well-refined structure⁹⁵. Furthermore, the E15R35 ternary mix exhibited similar nitrogen volume intrusion compared to CM but higher than the E05R15 and E10R25 ternary samples. This could be attributed to the enhanced porosity of the paste matrix, likely caused by unreacted silica in the samples with a high cement replacement percentage⁹⁶. The increased BET surface area observed in the E15R35 mix indicates microporosity resulting from agglomeration caused by the high replacement levels of RHA and EMS, both containing ultrafine particles. This agglomeration limits the effective participation of reactive silica in CSH formation, leading to reduced strength, as corroborated by SEM-EDS evidence of particle clustering. The BET surface area measurements provide valuable insights into the properties of the cementitious gel, particularly indicating the development of CSH phases. A higher BET surface area suggests an improved hydration process, contributing to the formation of additional CSH phases⁹⁷. A significant increase was recorded for ternary-based samples with various percentages compared to CM. This notable increase may be attributed to the increased EMS and RHA reactivity, enhancing the hydration process and creating denser phases of CSH. The E05R15 sample exhibited the highest BET surface area ($18.365\text{ m}^2/\text{g}$),

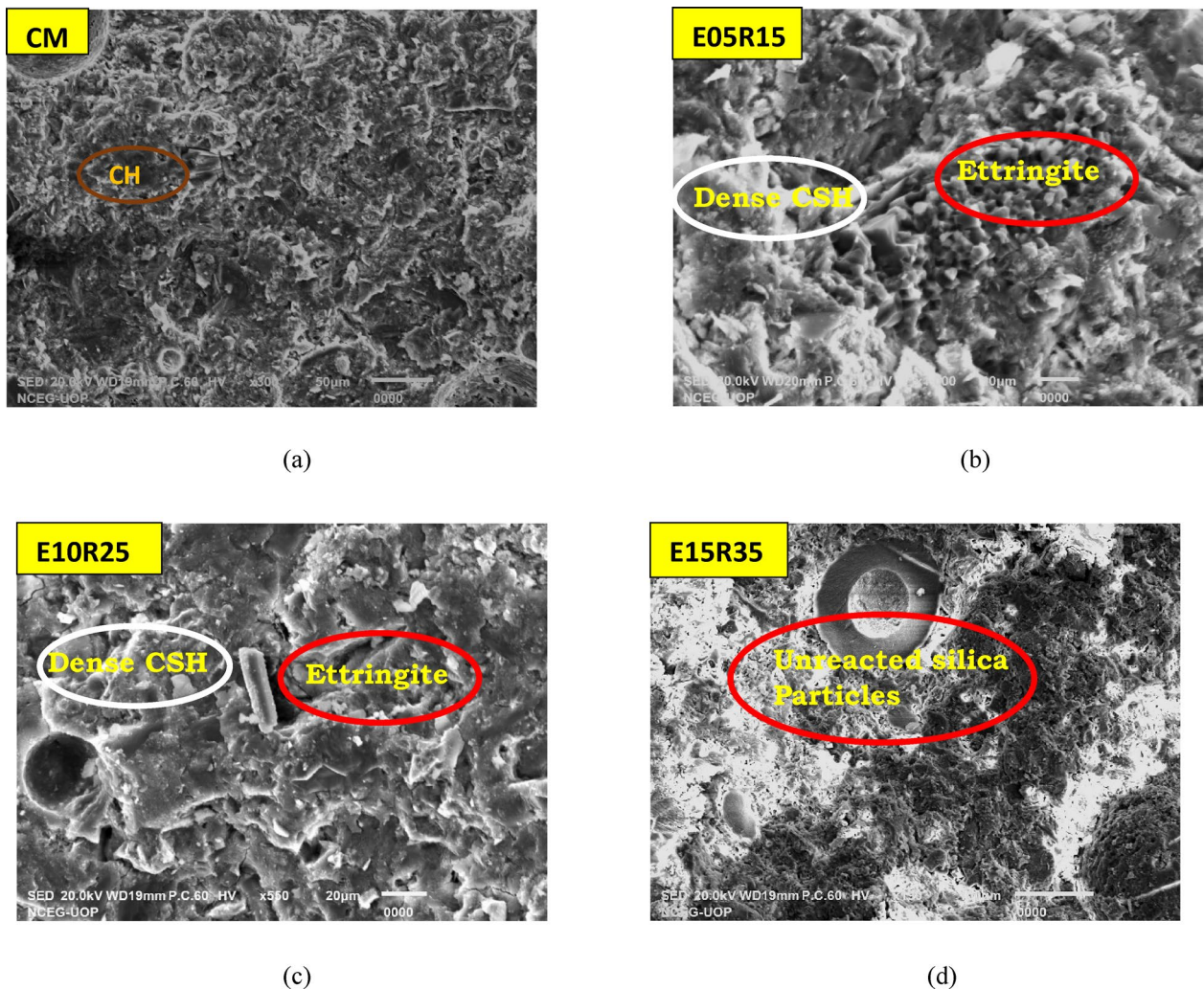


Fig. 15. SEM micrographs of control paste and ternary paste with different percentages after 28 days: (a) CM; (b) E05R15; (c) E10R25; (d) E15R35.

followed by E10R25 (17.879 m²/g) and E15R35 (15.633 m²/g). According to SEM, the slight reduction in BET surface area observed for both E10R25 and E15R35 could be attributed to agglomeration caused by oversaturation and inadequate dispersion, resulting in the development of less dense CSH phases. The decreased surface area confirms the presence of unreacted silica forming agglomerates within the paste matrix, causing oversaturation and disrupting the hydration process, leading to more porous structures⁹⁸.

Sustainability and practical implications

A comprehensive evaluation comparing concrete incorporating RHA and EMS with conventional silica fume-based mixes reinforces the innovative value of this study. Several studies have demonstrated the effectiveness of binary and ternary blends of RHA and SF in improving mechanical strength and durability. Zareei et al.³² reported that incorporating 20% RHA and 10% SF improved compressive strength and acid resistance while reducing water permeability and chloride ingress by 26% and 78%, respectively. Gupta et al.⁹⁹ observed strength increases of 4.12% with 10% RHA and 13.4% with 20% SF. Premkumar et al.³³ found that using 15% SF with 30% RHA enhanced compressive and flexural strength while reducing sorptivity and chloride penetration. Similarly, Xu et al.³⁴ showed compressive strength gains of 12.8% with 10% RHA and 17% with 10% SF, along with improved microstructure and durability. Sahoo et al.³⁵ demonstrated that mixes with 15% RHA and 15% SF enhanced acid resistance, reduced water penetration, and refined the microstructure. Amin et al.⁶⁵ achieved improved mechanical performance and reduced water absorption with a blend of 33% RHA and 7% SF, with lowered carbon emissions. Maharana et al.⁵⁰ also confirmed enhanced strength, durability, and cost efficiency using 10% RHA and 10% MS. In line with these findings, the present study reveals that the E05R15 and E10R25 mixes (high volume cement replacement) improved compressive strength by 11.56% and 5.98%, respectively, while achieving lower porosity and enhanced durability, confirmed by microstructural analysis showing dense CSH formation. The findings suggest that optimizing concrete with locally sourced supplementary cementitious materials (SCMs), such as RHA and EMS extracted from agricultural waste, can significantly reduce CO₂

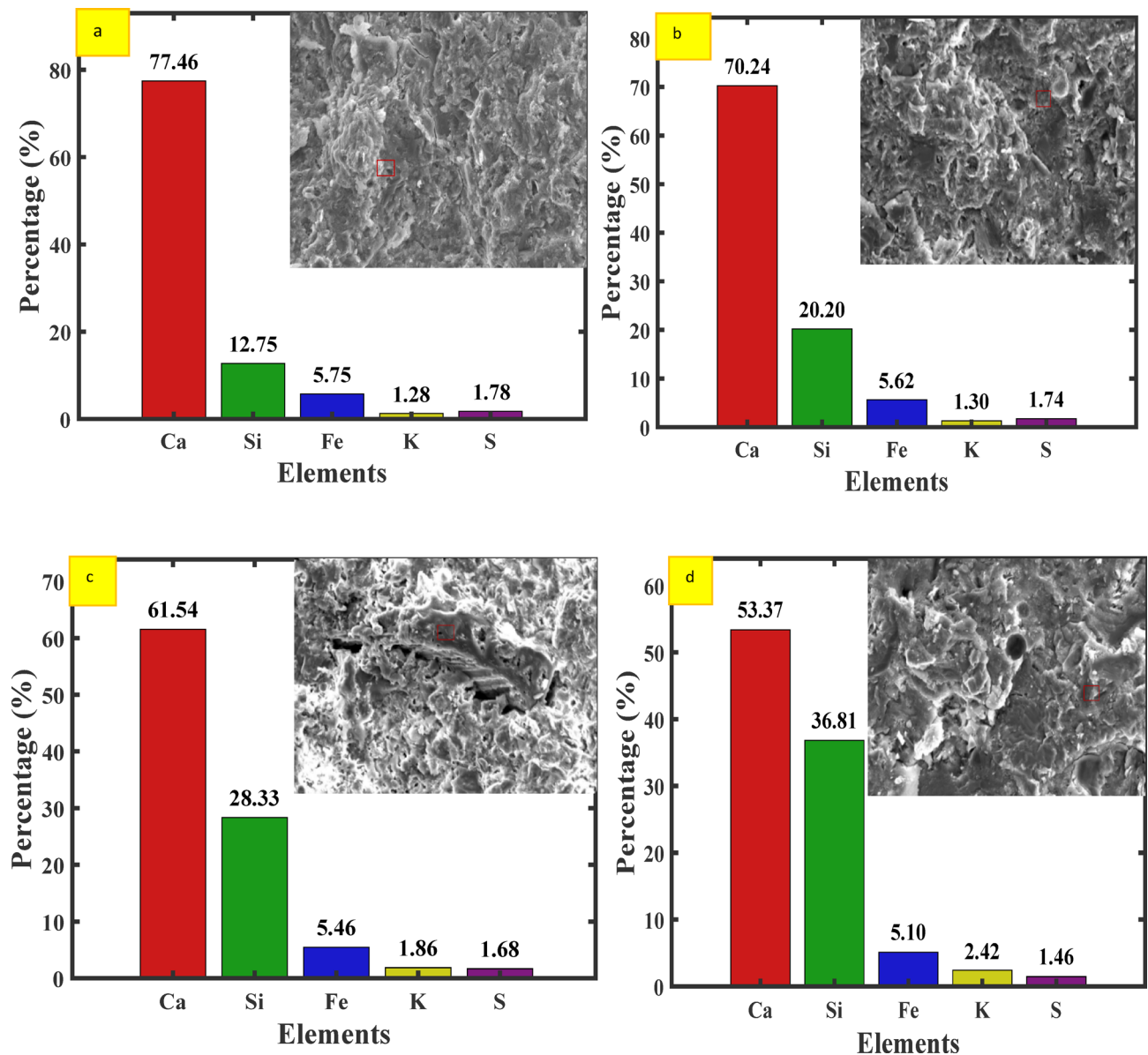


Fig. 16. EDS analysis for CM and ternary-based pastes with different percentages: (a) CM; (b) E05R15; (c) E10R25; (d) E15R35.

emissions and production costs without compromising strength and performance. The optimized mix design with 10% EMS and 25% RHA as cement replacements enhances mechanical properties and reduces reliance on conventional cement, offering a viable solution for the construction industry by utilizing local byproducts. While focusing on material characterization and mechanical behavior, the study highlights sustainability benefits such as reduced cement use, CO₂ reduction, agricultural waste valorization, and local material circularity. This makes the approach particularly viable for large-scale applications in regions like Pakistan, where sustainable construction is crucial to meet the challenges of rapid urbanization. These findings underscore the potential of industrial byproducts in developing green, high-performance concrete and inform future policies and practices toward low-carbon construction solutions.

Conclusions

This study examined the performance of RHA and microsilica chemically extracted from RHA in developing high-performance concrete. For the formation of high-performance and sustainable concrete mixes, different proportions of RHA (15%, 25%, and 35%) and EMS (5%, 10%, and 15%) were used as substitutes for cement. Finally, the combined effects of RHA and EMS as cement substitutes on mechanical, durability, and microstructural characteristics were studied.

The main conclusions drawn from the present experimental work are outlined below:

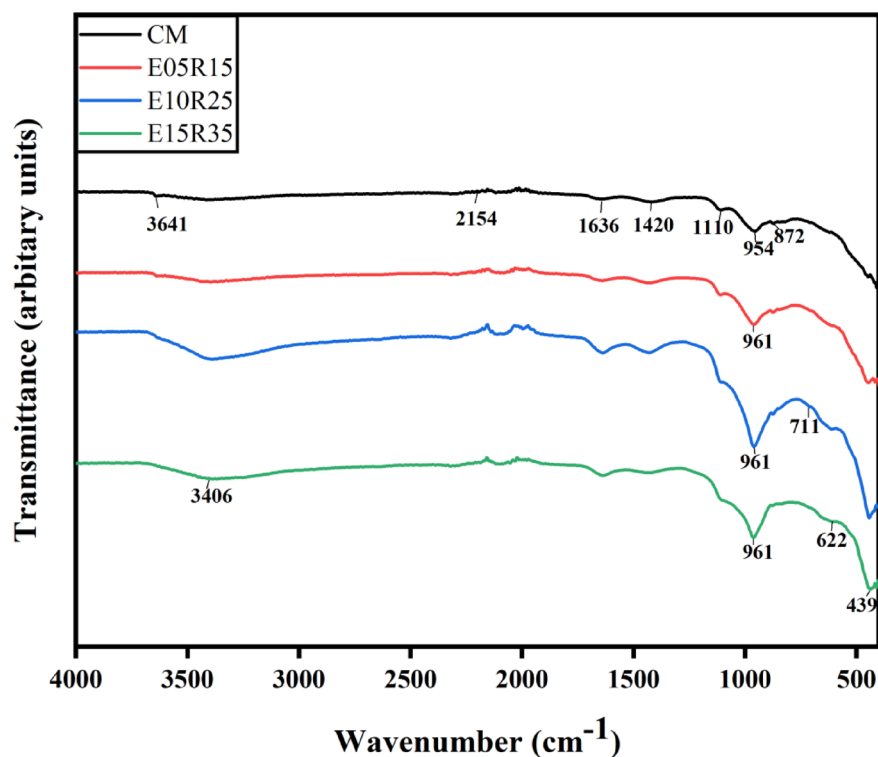


Fig. 17. Fourier-transform infrared spectroscopy spectra for CM and ternary samples with different percentages at 28 days.

Mix designation	BJH total pore volume (cm ³ /g)	BET surface area (m ² /g)
CM	0.006	6.491
E05R15	0.002	18.365
E10R25	0.005	17.879
E15R35	0.006	15.633

Table 8. BJH total pore volume and BET surface area of control and ternary paste samples containing different percentages of EMS and RHA at 28 days of age.

- The experimental findings demonstrate that the optimal 35% replacement threshold (25% RHA + 10% EMS) is governed by the synergistic material properties of the EMS and RHA. EMS's ultra-fine particle size (487.5 nm) and high silica purity (99.47%) contribute significantly to pore refinement and early-stage pozzolanic reactivity. In contrast, RHA, having a coarser average particle size (1273 nm) and slightly lower silica content (91.03%), supports long-term strength development. The notable differences in particle size distribution and specific gravity between EMS and RHA enable a favorable packing effect, where finer EMS particles effectively fill the interstitial voids between coarser RHA particles, thereby enhancing matrix densification. Furthermore, the high surface area of EMS (36.58 m²/g) compensates for the lower surface area of RHA (0.360 m²/g), ensuring sufficient CSH gel formation up to the 35% replacement level. However, beyond this limit, the performance decline in the high replacement E15R35 mix was attributed to the accumulation of unreacted silica forming agglomerates, affecting the hydration process and resulting in weak hydration products.
- Compressive strength results revealed that ternary mixes (E05R15 and E10R25) resulted in higher compressive strength, whereas (E15R35) caused a decrease in compressive strength than the CM. Moreover, the results indicate that compressive strength increased significantly for samples having (E05R15) and (E10R25) by 11.56% and 5.98%, while (E15R35) decreased by 12.55% when compared with the CM after 28 days. Similar to the compressive strength, the splitting tensile strength was also markedly increased for the (E05R15 and E10R25) and decreased for the (E15R35) compared to the CM. Furthermore, the proposed equation by Zain et al. and CEB-FIB provides the most accurate prediction of split tensile strength.
- Water absorption and apparent porosity decreased as the percentage of cement replacement increased up to 35% (E05R15 and E10R25). However, WA and AP increased when the cement replacement percentage was increased by 50% (E15R35) compared to the CM. Additionally, the water penetration decreased for concrete mixes containing E05R15 and E10R25 but increased for those containing E15R35.

- SEM analysis shows improvements in the microstructure of E05R15 and E10R25 mixes. The results of the XRD analysis showed a significant reduction in the Portlandite peak in mixes containing E05R15 and E10R25, which is attributed to the better reactivity of both RHA and EMS. Furthermore, SEM-EDS analysis also showed the development of CSH phases with lower Ca/Si ratios, possibly due to the higher reactivity of ultra-fine silica in these mixes, resulting in high-density CSH phases.
- FTIR analysis results indicate a peak between 954 cm^{-1} and 961 cm^{-1} , which is typically attributed to the CSH phase in these samples. A shift in peak intensity was observed in all ternary samples compared to control samples, indicating the development of denser CSH phases. The FTIR peak at 3641 cm^{-1} confirmed the presence of OH groups, indicating the portlandite phase. Notably, the E05R15 sample showed a modest Portlandite peak, while the E10R25 and E15R35 substituted samples showed considerably smaller or completely disappeared peaks, indicating higher pozzolanic activity of EMS and RHA present in these mixes.
- N_2 adsorption results revealed that the BET surface area increased while intruded pore volume decreased in ternary mixtures containing E05R15 and E10R25. This could be explained by the better reactivity of both RHA and EMS, which resulted in the formation of more CSH gel and a decrease in the porosity. In contrast, the ternary mix with E15R35 displayed a low BET surface area and high pore volume due to the presence of unreacted phases and agglomeration of EMS when used at high volume in this mixture.

The feasibility of using EMS combined with RHA as a partial substitution of cement in concrete revealed great potential in terms of enhanced mechanical strength, durability properties, and improved microstructure. However, researchers should consider high shear mixing and ultrasonic dispersion techniques in the future, which could result in effective utilization of EMS along with RHA at high volume and with improved performance. Further studies should also aim to improve water penetration resistance while maintaining these key properties. Advanced techniques such as isothermal calorimetry and thermogravimetric analysis are recommended to quantify hydration behavior. Moreover, comprehensive life cycle assessments should be conducted to evaluate the environmental impact of EMS production, including energy consumption, CO_2 emissions, and chemical usage. Pilot-scale trials are also essential to ensure process consistency across diverse RH sources and to refine reagent recovery strategies for industrial-scale implementation.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions

Muhammad Fahad Ullah, Hesheng Tang, Arshad Ullah, and Kaffayatullah khan were responsible for the conceptualization and design of the study. Muhammad Fahad Ullah and Hesheng Tang conducted data collection and analysis. Muhammad Fahad Ullah and Arshad Ullah prepared the initial draft of the manuscript. Hesheng Tang and Kaffayatullah khan provided critical reviews and revisions to the manuscript. Hesheng Tang secured funding for the article processing charges. All authors reviewed and approved the final version of the manuscript for submission.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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